Supplemental Digital Content 5. Continuous outcomes, general quality of life, psychosocial well-being, sexual well-being, patient satisfaction with aesthetics, and patient satisfaction with outcome

Study, Year, PMID, Country	Design	Overall RoB	Outcome	Outcome Measurement	Time Point	Subgroup	IBR, N	IBR, Mean (SD)	AR, N	AR, Mean (SD)	AR vs. IBR, Effect Size (95% CI)	P Value
Tallroth, 2020, 33436336, Sweden	RCT	Moderate	Psychosocial WB	BREAST-Q: Psychosocial WB (0-100)	5.3 y	All	28	78.8 (20.1)	42	79.1 (21.5)	MD 0.3 (−6.7, 7.3)	0.93
Eltahir, 2015, 25539295, Netherlands	NRCS	Moderate	Psychosocial WB	BREAST-Q: Psychosocial WB (0-100)	2.2 y	All	45	77.2 (18.1)	47	74.0 (17.8)	adjMD 4.6 (−2.8, 12.0)	0.22
	NRCS	Moderate	Psychosocial WB	SF-36: Social functioning (0-100)	2.2 y	All	NR	NR	NR	NR	adjMD −1.21 (−8.44, 6.02)	0.74
Kouwenberg, 2020, 32590633, Netherlands	NRCS	Moderate	Psychosocial WB	BREAST-Q: Psychosocial WB (0-100)	>6 mo	All	296	71.6 (20.2)	179	75.8 (19.5)	NR	<0.05
	NRCS	Moderate	Psychosocial WB	EORTC QLQC30: Social function (0-100)	>6 mo	All	296	87.5 (27.2)	179	88.0 (17.1)	NR	NS
	NRCS	Moderate	Psychosocial WB	EORTC QLQC30: Emotional function (0-100)	>6 mo	All	296	85.0 (23.3)	179	87.0 (17.1)	NR	NS
	NRCS	Moderate	Psychosocial WB	EORTC QLQC30: Cognitive function (0-100)	>6 mo	All	296	85.0 (23.7)	179	83.7 (21.5)	NR	NS
	NRCS	Moderate	Psychosocial WB	EORTC QLQC30: Role function (0-100)	>6 mo	All	296	86.0 (28.1)	179	84.0 (21.8)	NR	<0.05

Study, Year, PMID, Country	Design	Overall RoB	Outcome	Outcome Measurement	Time Point	Subgroup	IBR, N	IBR, Mean (SD)	AR, N	AR, Mean (SD)	AR vs. IBR, Effect Size (95% CI)	P Value
Kulkarni, 2017, 28713853, US & Canada	NRCS	Moderate	Psychosocial WB	BREAST-Q: Psychosocial WB (0-100)	1 y	All	791	71.8 (19)	386	74.7 (19.2)	adjMD 3.70 (0.73, 6.76)	0.015
	NRCS	Moderate	Psychosocial WB	BREAST-Q: Psychosocial WB (0-100)	2 y	All	1490	74.5 (18.9)	523	75.8 (19)	adjMD 3.27 (1.25, 5.29)	NR
	NRCS	Moderate	Psychosocial WB	BREAST-Q: Psychosocial WB (0-100)	2 y	Unilateral	600	74.6 (18.7)	317	76.8 (18.9)	adjMD 3.84 (NR)	NR
	NRCS	Moderate	Psychosocial WB	BREAST-Q: Psychosocial WB (0-100)	2 y	Bilateral	994	74.5 (19)	224	73.4 (19)	adjMD 0.91 (NR)	NR
	NRCS	Moderate	Psychosocial WB	PROMIS: Anxiety (0-100)	1 y	All	775	49.7 (9.4)	383	50.4 (9.6)	adjMD 0.70 (−0.75, 2.08)	0.36
	NRCS	Moderate	Psychosocial WB	PROMIS: Depression (0-100)	1 y	All	776	47.3 (8)	385	47.9 (8.2)	adjMD 0.40 (−0.70, 1.45)	0.50
Roth, 2007, 17413877, US	NRCS	High	Psychosocial WB	FACT-B: Social/ family WB (0-28)	2 y	All	35	19.3 (NR)	55	20.3 (NR)	NR	0.24
	NRCS	High	Psychosocial WB	SF-36: Social functioning (0-100)	2 y	All	35	87.9 (NR)	55	87.7 (NR)	NR	≥0.05

Study, Year, PMID, Country	Design	Overall RoB	Outcome	Outcome Measurement	Time Point	Subgroup	IBR, N	IBR, Mean (SD)	AR, N	AR, Mean (SD)	AR vs. IBR, Effect Size (95% CI)	P Value
Tallroth, 2020, 33436336, Sweden	RCT	Moderate	Sexual WB	BREAST-Q: Sexual well-being (0-100)	5.3 y	All	28	58.4 (23.1)	42	67.1 (28.1)	MD 8.7 (0.2, 17.2)	0.046
Kouwenberg, 2020, 32590633, Netherlands	NRCS	Moderate	Sexual WB	BREAST-Q: Sexual well-being (0-100)	>6 mo	All	296	56.4 (30.7)	179	63.3 (30.4)	NR	<0.05
	NRCS	Moderate	Sexual WB	EORTC QLQBR23: Sexual functioning (0-100)	>6 mo	All	296	33.4 (29.4)	179	32.0 (27.3)	NR	<0.05
	NRCS	Moderate	Sexual WB	EORTC QLQBR23: Sexual enjoyment (0-100)	>6 mo	All	296	63.8 (32.5)	179	64.2 (33.8)	NR	NS
Eltahir, 2015, 25539295, Netherlands	NRCS	Moderate	Sexual WB	BREAST-Q: Sexual well-being (0-100)	2.2 y	All	45	61.14 (24.17)	47	60.89 (20.82)	adjMD 6.44 (−3.56, 16.5)	0.20
Kulkarni, 2017, 28713853, US & Canada	NRCS	Moderate	Sexual WB	BREAST-Q: Sexual WB (0-100)	1 y	All	756	53 (21.1)	370	55.4 (19.8)	adjMD 4.50 (1.52, 7.48)	0.003
	NRCS	Moderate	Sexual WB	BREAST-Q: Sexual WB (0-100)	2 y	All	1490	53.9 (21.3)	523	57.1 (21.7)	adjMD 5.53 (−2.95, 8.11)	NR
	NRCS	Moderate	Sexual WB	BREAST-Q: Sexual WB (0-100)	2 y	Unilateral	600	52.8 (20.5)	317	58.9 (20.6)	adjMD 11.42 (NR)	<0.001

Study, Year, PMID, Country	Design	Overall RoB	Outcome	Outcome Measurement	Time Point	Subgroup	IBR, N	IBR, Mean (SD)	AR, N	AR, Mean (SD)	AR vs. IBR, Effect Size (95% CI)	P Value
	NRCS	Moderate	Sexual WB	BREAST-Q: Sexual WB (0-100)	2 y	Bilateral	994	54.7 (21.5)	214	54.4 (23)	adjMD 4.2 (NR)	<0.001
Tallroth, 2020, 33436336, Sweden	RCT	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with breast (0-100)	5.3 y	All	28	63.4 (11.8)	42	72.1 (17.7)	MD 8.7 (3.8, 13.6)	0.001
	RCT	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with nipples (0-100)	5.3 y	All	28	65.4 (21.8)	42	67.7 (24.9)	MD 2.3 (−5.5, 10.1)	0.56
Brito, 2020, No PMID, Portugal	NRCS	High	Satisfaction with breast	BREAST-Q: Satisfaction with breast (0-100)	NR	All	68	56.3 (17.1)	111	64.1 (17.1)	NR	0.004
Eltahir, 2015, 25539295, Netherlands	NRCS	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with breast (0-100)	2.2 y	All	45	65.51 (17.55)	47	75.19 (17.09)	adjMD 8.16 (1.18, 15.2)	0.023
	NRCS	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with nipples (0-100)	2.2 y	All	45	63.62 (33.99)	47	65.31 (27.82)	adjMD 1.70 (−14.2, 17.6)	0.83
Kouwenberg, 2020, 32590633, Netherlands	NRCS	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with breast (0-100)	>6 mo	All	296	59.4 (19.3)	179	71.3 (17.7)	NR	<0.05
	NRCS	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with nipples (0-100)	>6 mo	All	296	55.0 (48.7)	179	63.0 (29.0)	NR	<0.05
Kulkarni, 2017, 28713853,	NRCS	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with breast (0-100)	1 y	All	795	64.0 (16.8)	388	67.8 (17.2)	adjMD 6.30 (3.41, 9.09)	<0.001

Study, Year, PMID, Country	Design	Overall RoB	Outcome	Outcome Measurement	Time Point	Subgroup	IBR, N	IBR, Mean (SD)	AR, N	AR, Mean (SD)	AR vs. IBR, Effect Size (95% CI)	P Value
US & Canada	NRCS	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with breast (0-100)	2 y	All	1490	64.2 (18)	523	68.5 (18.3)	adjMD 7.94 (5.68, 10.2)	NR
	NRCS	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with breast (0-100)	2 y	Unilateral	600	61.2 (18)	317	68.3 (18.1)	adjMD 9.85 (NR)	0.001
	NRCS	Moderate	Satisfaction with breast	BREAST-Q: Satisfaction with breast (0-100)	2 y	Bilateral	994	66.1 (17.7)	214	68.9 (18.6)	adjMD 5.13 (2.07, 8.17)*	0.001
Tallroth, 2020, 33436336, Sweden	RCT	Moderate	Satisfaction with outcome	BREAST-Q: Satisfaction with outcome (0-100)	5.3 y	All	28	79.4 (14.2)	42	82.3 (21.4)	MD 2.9 (−3.1, 8.9)	0.34
Eltahir, 2015, 25539295, Netherlands	NRCS	Moderate	Satisfaction with outcome	BREAST-Q: Satisfaction with outcome (0-100)	2.2 y	All	45	74.5 (19.0)	47	81.8 (18.7)	adjMD 4.9 (−3.1, 12.9)	0.23
Kouwenberg, 2020, 32590633, Netherlands	NRCS	Moderate	Satisfaction with outcome	BREAST-Q: Satisfaction with outcome (0-100)	>6 mo	All	296	66.4 (23.7)	179	75.8 (22.2)	NR	<0.05
Kouwenberg, 2019, 30270015, Netherlands	NRCS	Moderate	General QoL	EQ-5D-5L utilities score (0-1)	NR	All	67	0.85 (0.18)	67	0.87 (0.14)	NR	0.7
Kouwenberg, 2020, 32590633, Netherlands	NRCS	Moderate	General QoL	EQ-5D-5L utilities score (0-1)	>6 mo	All	296	0.85 (0.30)	179	0.85 (0.20)	NR	NS
	NRCS	Moderate	General QoL	EQ-5D-5L VAS (0-100)	>6 mo	All	296	77.6 (18.4)	179	79.2 (16.7)	NR	NS

Study, Year, PMID, Country	Design	Overall RoB	Outcome	Outcome Measurement	Time Point	Subgroup	IBR, N	IBR, Mean (SD)	AR, N	AR, Mean (SD)	AR vs. IBR, Effect Size (95% CI)	P Value
	NRCS	Moderate	General QoL	EORTC QLQC30: Global health status (0-100)	>6 mo	All	296	80.2 (18.4)	179	81.4 (14.7)	NR	NS
Roth, 2007, 17413877, US	NRCS	High	General QoL	FACT-B: Functional well-being (0-28)	2 y	All	35	23.3 (NR)	55	24.1 (NR)	NR	NS
	NRCS	High	General QoL	SF-36: Role emotional (0-100)	2 y	All	35	86.7 (NR)	55	86.1 (NR)	NR	NS
	NRCS	High	General QoL	SF-36: Vitality (0-100)	2 y	All	35	65.7 (NR)	55	62.4 (NR)	NR	NS
	NRCS	High	General QoL	SF-36: General mental health (0-100)	2 y	All	35	77.6 (NR)	55	77 (NR)	NR	NS
	NRCS	High	General QoL	Body Image (9-45)	2 y	All	35	32.5 (NR)	55	35.3 (NR)	NR	NS

Abbreviations: adj = adjusted, CI = confidence interval, EQ-5D = EuroQoL 5D, EORTC = European Organization for Research and Treatment of Cancer , QoL = quality of life, MD = mean difference, mo = months, NMD = net mean difference, NR = not reported, NRCS = nonrandomized comparative study, PMID = PubMed identifier, RCT = randomized controlled trial, RoB = risk of bias, SD = standard deviation, SF-36 = Short Form-36, VAS = Visual Analog Scale, WB = well-being, y = years.

* Confidence interval calculated based on the reported P value.